Minglu Zhao

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ACADEMIC BACKGROUND

University of California, Los Angeles (UCLA)
Ph.D. Candidate in Machine Learning Sep 2021 - 2026 (Expected)
GPA: 4.0 / 4.0; Co-advised by Prof. Ying Nian Wu and Prof. Tao Gao

B.S. Statistics, B.S. Cognitive Science, minor in Mathematics Sep 2017 - Dec 2020
GPA: 3.9 / 4.0; *Magna Cum Laude*

RESEARCH INTERESTS

Language model reasoning, Decision making, Representation learning

SELECTED PUBLICATIONS

(* denotes equal contributions)

Deqian Kong*, **Minglu Zhao***, Aoyang Qin*, Bo Pang, Chenxin Tao, David Hartmann, Edouardo Honig, Dehong Xu, Amit Kumar, Matt Sarte, Chuan Li, Jianwen Xie, Ying Nian Wu. "Inference-Time Rethinking with Latent Thought Vectors for Math Reasoning." ***ICLR*** *Workshop on Latent & Implicit Thinking*, **2026**.

Minglu Zhao*, Dehong Xu*, Deqian Kong*, Wen-Hao Zhang, Ying Nian Wu. "Place Cells as Multi-Scale Position Embeddings: Random Walk Transition Kernels for Path Planning." *Conference on Neural Information Processing Systems (**NeurIPS**)*, **2025**.

Donghun Noh*, Deqian Kong*, **Minglu Zhao**, Andrew Lizarraga, Jianwen Xie, Ying Nian Wu, Dennis Hong. "Latent Adaptive Planner for Dynamic Manipulation." *Conference on Robot Learning (**CoRL**)*, **2025**.

Deqian Kong*, **Minglu Zhao***, Dehong Xu*, Bo Pang, Shu Wang, Edouardo Honig, Zhangzhang Si, Chuan Li, Jianwen Xie, Sirui Xie, Ying Nian Wu. "Latent Thought Models with Variational Bayes Inference-Time Computation." *International Conference on Machine Learning (**ICML**)*, **2025**.

Minglu Zhao, Dehong Xu, Wenhao Zhang, Ying Nian Wu, "A Minimalistic Representation Model for Head Direction System." *Annual Conference of the Cognitive Science Society (**CogSci**)*, **2025**.

Qian Long*, Ruoyan Li*, **Minglu Zhao*,** Tao Gao, Demetri Terzopoulos. "Inverse Attention Agent in Multi-Agent System." *International Conference on Learning Representations (**ICLR**)*, **2025**.

Deqian Kong*, Dehong Xu*, **Minglu Zhao***, Bo Pang, Jianwen Xie, Andrew Lizarraga, Yuhao Huang, Sirui Xie, Ying Nian Wu. "Latent Plan Transformer: Planning as Latent Variable Inference." *Conference on Neural Information Processing Systems (**NeurIPS**)*, **2024**.

Shaozhe Cheng, **Minglu Zhao**, Ning Tang, Yang Zhao, Jifan Zhou, Mowei Shen, and Tao Gao. "Intention Beyond Desire: Spontaneous Intentional Commitment Regulates Conflicting Desires." ***Cognition*** *238 (2023): 105513.*

Minglu Zhao, Ning Tang, Annya L. Dahmani, Yixin Zhu, Federico Rossano, and Tao Gao. "Sharing Rewards Undermines Coordinated Hunting." *Journal of Computational Biology 29, no. 9 (2022): 1022-1030.*

Ning Tang, Siyi Gong, **Minglu Zhao**, Chenya Gu, Jifan Zhou, Mowei Shen, and Tao Gao. "Exploring an Imagined "We" in Human Collective Hunting: Joint Commitment within Shared Intentionality." *Annual Conference of the Cognitive Science Society (**CogSci**)*, **2022**.

Stephanie Stacy, Chenfei Li, **Minglu Zhao**, Yiling Yun, Qingyi Zhao, Max Kleiman-Weiner, and Tao Gao. "Modeling Communication to Coordinate Perspectives in Cooperation." *Annual Conference of the Cognitive Science Society (**CogSci**)*, **2021**.

RESEARCH EXPERIENCE

Meta Superintelligence Labs (MSL) Jun 2026 - Aug 2026
Research scientist intern, Advisor: Michael Tontchev, Justin Zhao

- Focused on AI safety evaluation; proposed and developed an automated multi-turn attack mining framework and conducted the first systematic study across internal and public benchmarks, uncovering model-specific vulnerabilities in both safety violation and false-refusal behaviors. Research paper in progress.
- Improved Meta's multi-turn safety benchmark generation pipeline, increasing the internal benchmark violation rate by 6× through a generalizable generation and filtering design.

Meta Research Jun 2025 - Sep 2025
Research scientist intern, Advisor: Dr. Yunbo Ouyang

- Proposed and implemented a novel reranking framework for the internal recommendation system, resulting in significant improvements across key evaluation metrics.
- Designed and conducted systematic online studies over 10+ model variants, leading to two high-performing candidates selected for production launch.

Natera, Inc. Feb 2025 - June 2026
Research scientist intern, Advisor: Dr. Zijun Frank Zhang

- Designed and pretrained an efficient foundation model for multimodal genomics data, launched as part of Natera's large-scale genomics AI platform trained on one of the largest datasets in oncology.
- Developed scalable data preprocessing pipelines to unify heterogeneous genomic, clinical, and imaging modalities.

GE Research Jun 2023 - Sep 2023
Research scientist intern, Advisor: Dr. Peter Tu

- Designed a real-time video understanding framework that extracts and classifies predicates and actions from real-world video, enabling automated plan recognition and high-level goal inference in dynamic settings.

UCLA, Department of Statistics Sep 2021 - present
Research assistant, Advisors: Prof. Ying Nian Wu and Prof. Tao Gao

- Language modeling with latent thought representations
  - Proposed *Latent-Thought Language Models (LTMs)*, incorporating explicit latent thought vectors and a dual-rate variational Bayes framework. This approach achieves superior sample efficiency, emergent in-context reasoning,

and outperforms conventional autoregressive models in zero-shot and few-shot language modeling.
  - Proposed *Inference-Time Rethinking*, a latent-variable framework that performs iterative test-time optimization over continuous thought vectors via a Gibbs-style generate–reflect loop, enabling self-correction and substantially improving mathematical reasoning performance

- Decision-making and robotics as latent plan inference
  - Planning as latent variable inference: Developed the *Latent Plan Transformer*, an unsupervised solution to decision-making via sequence modeling by inferring a latent variable from a target return to guide policy execution as a plan.
  - Iterative latent plan refinement for robotics control: developed a real-time replanning method using a few Langevin steps at each timestep, enhancing robustness under uncertainty while remaining computationally efficient.

- Multi-agent RL inspired by Theory-of-Mind
  - Proposed *Imagined We*, a Bayesian Theory-of-mind approach that bridges centralized and decentralized multi-agent systems via a unified "We" perspective, enabling superior cooperation and more human-like behavior.
  - Developed the *Inverse Attention Agents* that leverage an attention mechanism to dynamically infer and adjust to other agents' goals in multi-agent environments, enhancing cooperation, competition, and human-agent interactions.

ORAL PRESENTATIONS

- *Conference of the Cognitive Science Society (CogSci)*, 2022 **(Oral Presentation)**
  Exploring an Imagined "We" in Collective Hunting: Joint Commitment within Shared Intentionality
- *RSS Workshop on Social Intelligence in Humans and Robots*, 2022 **(Spotlight)**
  Exploring an Imagined "We" in Collective Hunting: Joint Commitment within Shared Intentionality
- *Conference of the Cognitive Science Society (CogSci)*, 2021 **(Oral Presentation)**
  Modeling Animal Coordinated Hunting with Reinforcement Learning
- *ICRA Workshop on Social Intelligence in Humans and Robots,* 2021 **(Spotlight)**
  Modeling Communication to Coordinate Perspectives in Cooperation
- *ICRA Workshop on Social Intelligence in Humans and Robots,* 2021 **(Spotlight)**
  Modeling Animal Coordinated Hunting with Reinforcement Learning

AWARDS

Graduate Student Fellowship, UCLA	2021 - 2026
Doctoral Student Travel Award, UCLA	2021 - 2026
UCLA Dean's Honor List for Superior Academic Achievement	2017 - 2021

PROFESSIONAL SERVICES

Peer-reviewed Journals and Conferences

Transactions on Machine Learning Research (TMLR)
International Conference on Machine Learning (ICML)
International Conference on Learning Representations (ICLR)
Conference on Neural Information Processing Systems (NeurIPS)
International Conference on Artificial Intelligence and Statistics (AISTATS)
Annual AAAI Conference on Artificial Intelligence (AAAI)
Annual Conference of the Cognitive Science Society (CogSci)